THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A letter is chosen at random from the word RAJASTHAN. The probability that it is a vowel is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{6}$

2. Three coins are tossed together. The probability of getting all tails is: 1

(a) $\frac{1}{8}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{7}{8}$

3. A bag has 6 red, 2 black and 7 white balls. The probability of drawing a ball that is not black is: 1

(a) $\frac{2}{15}$ **(b)** $\frac{13}{15}$ **(c)** $\frac{14}{15}$ **(d)** $\frac{13}{16}$

4. If $P(E) = 0.4$, then $P(\text{not } E) \times 100$ equals: 1

(a) 59 **(b)** 62 **(c)** 61 **(d)** 60

5. A fair die is thrown once. The probability of getting a number between 2 and 5 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{6}$

6. Outcomes that have the same chance of occurring are called: 1

(a) equally likely **(b)** certain **(c)** impossible **(d)** complementary

7. A die is thrown 100 times with frequencies of $1, \ldots, 6$ being 18, 12, 20, 15, 19, 16. The experimental probability of getting 2 is: 1

(a) $\frac{22}{25}$ **(b)** $\frac{3}{25}$ **(c)** $\frac{4}{25}$ **(d)** $\frac{3}{26}$

8. In a survey of 250 families, 75 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{2}{5}$ **(b)** $\frac{3}{10}$ **(c)** $\frac{7}{10}$ **(d)** $\frac{3}{11}$

9. A letter is chosen at random from the word PROBABILITY. The probability that it is a vowel is: 1

(a) $\frac{7}{11}$ **(b)** $\frac{5}{11}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{1}{3}$

10. Three coins are tossed together. The probability of getting at least two heads is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{4}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A letter is chosen at random from the word RAJASTHAN. Find the probability that it is a vowel. 2

12. Three coins are tossed together. Find the probability of getting at least two heads. 2

13. A die was thrown 100 times with frequencies of $1, \ldots, 6$ being 21, 17, 14, 20, 13, 15. Find the experimental probability of getting an even number. 2

14. A letter is chosen at random from the word PROBABILITY. Find the probability that it is a consonant. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains 5 red, 5 black and 5 white balls. A ball is drawn at random. Find the probability that it is — 3
(i) red,
(ii) not red,
(iii) black or white.

16. In a survey of 120 students, 45 take the bus, 30 cycle and the rest walk. Find the probability that a randomly chosen student — 3
(i) walks,
(ii) does not take the bus.

17. Two coins are tossed 80 times: two heads occurred 18 times, one head 42 times and no head 20 times. Find the experimental probability of each of the three events. — 3

OR

A bag contains 7 red, 9 black and 4 white balls. A ball is drawn at random. Find the probability that it is — 3
(i) red,
(ii) not red,
(iii) black or white.

18. In a survey of 300 students, 120 take the bus, 90 cycle and the rest walk. Find the probability that a randomly chosen student — 3
(i) walks,
(ii) does not take the bus.

OR

Two coins are tossed 160 times: two heads occurred 38 times, one head 78 times and no head 44 times. Find the experimental probability of each of the three events. — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A die was thrown 100 times; the number of times $1, \ldots, 6$ appeared was 21, 17, 14, 20, 13, 15. Find the experimental probability of — 5
(i) a prime number,
(ii) a number greater than 4,
(iii) a number less than 3, and check that all six relative frequencies add to 1.

OR

A letter is chosen at random from the word EXAMINATION. Find the probability that it is — 5
(i) a vowel,
(ii) the letter N,
(iii) a letter from the first half of the alphabet (A to M).

20. A die was thrown 120 times; the number of times $1, \ldots, 6$ appeared was 18, 22, 19, 21, 17, 23. Find the experimental probability of — 5
(i) a prime number,
(ii) a number greater than 4,
(iii) a number less than 3, and check that all six relative frequencies add to 1.

OR

A letter is chosen at random from the word PROBABILITY. Find the probability that it is — 5
(i) a vowel,
(ii) the letter B,
(iii) a letter from the first half of the alphabet (A to M).

END OF QUESTION PAPER — Total: 40 Marks

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